

An aerial photograph of Singapore's skyline at sunset. The Marina Bay Sands hotel is prominent on the left, with its distinctive three towers and a skybridge. The city's skyscrapers are visible in the background, and the water of Marina Bay is in the foreground. The sky is a mix of orange, pink, and blue.

# Market Dynamics of Singapore's Banking Sector

**Team: Banklytics**

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## Motivation & Background

Singapore's major banks are DBS, OCBC and UOB which are the backbone of the economy.

Understanding the movement of their stock prices can help understand and predict market trends, macroeconomic factors, and global events.

Predict the stock price of major banks using deep learning-based models



## Project Objectives

1. Find the perfect combination of technical indicators.
2. Integrating sentimental scores
3. Use log returns for the modeling target.
4. Develop a LSTM model architecture for prediction.
5. Evaluate the model performance using walk-forward expanding-window.
6. Directional accuracy as a primary evaluation metric.



## Data Overview: Yfinance Data

**Data Collection:** Using yFinance API in Python

**STOCK NAME:** U11.SI for UOB, D05.SI for DBS and O39.SI and OCBC.

**Coverage:** 2024–2025

**Focus Variable:** Open, High, Low, Close, Volume(OHLCV)



## Data Overview: Sentiment data

### Data Collection: Using GDELT API In R

| bank      | text                                                             | published           | source                       | url                                                             |
|-----------|------------------------------------------------------------------|---------------------|------------------------------|-----------------------------------------------------------------|
| OCBC Bank | Asia shares get 2024 to steady start with busy data calendar...  | 2024-01-02 03:00:00 | channelnewsasia.com          | https://www.channelnewsasia.com/business/asia-shares-get...     |
| OCBC Bank | Asia shares : Asian shares get 2024 to steady start with busy... | 2024-01-02 03:00:00 | economictimes.indiatimes.com | https://economictimes.indiatimes.com/markets/stocks/news...     |
| OCBC Bank | Asia shares get 2024 to steady start with busy data calendar...  | 2024-01-02 03:45:00 | marketscreener.com           | https://www.marketscreener.com/news/latest/Asia-shares-g...     |
| OCBC Bank | Asia shares get 2024 to steady start with busy data calendar...  | 2024-01-02 04:00:00 | finance.yahoo.com            | https://finance.yahoo.com/news/asia-shares-2024-steady-st...    |
| OCBC Bank | Singapore Q4 GDP growth speeds up                                | 2024-01-02 04:30:00 | asiaone.com                  | https://www.asiaone.com/money/singapores-q4-gdp-growt...        |
| OCBC Bank | Asia shares get 2024 to steady start with busy data calendar...  | 2024-01-02 05:00:00 | zawya.com                    | https://www.zawya.com/en/markets/equities/asia-shares-ge...     |
| OCBC Bank | Asia shares kick off 2024 on steady footing ; bitcoin surges     | 2024-01-02 08:30:00 | business.inquirer.net        | https://business.inquirer.net/439319/asia-shares-kick-off-20... |
| OCBC Bank | Asia shares get 2024 to steady start with busy data calendar...  | 2024-01-02 09:15:00 | brecorder.com                | https://www.brecorder.com/news/40281645/asia-shares-get...      |
| UOB Bank  | IPO Multi Spunindo ( MSJA ) di Harga Rp300 , Cek Rencana...      | 2024-01-03 09:00:00 | market.bisnis.com            | https://market.bisnis.com/read/20240103/192/1729141/ipo...      |
| UOB Bank  | European Midday Briefing : Stocks Drift Lower as Fed Minut...    | 2024-01-03 11:30:00 | morningstar.com              | https://www.morningstar.com/news/dow-jones/202401031...         |

### Sentiment Scoring: Using Vader to generate sentiment scores for the text

| bank      | date       | n_articles | avg_compound | sd_compound | pct_positive | pct_negative |
|-----------|------------|------------|--------------|-------------|--------------|--------------|
| OCBC Bank | 2024-01-02 | 8          | 0.335625000  | 0.08297665  | 1.0000000    | 0.00000000   |
| OCBC Bank | 2024-01-03 | 1          | 0.848000000  | NA          | 1.0000000    | 0.00000000   |
| UOB Bank  | 2024-01-03 | 3          | -0.076666667 | 0.09152777  | 0.0000000    | 0.66666667   |
| OCBC Bank | 2024-01-04 | 9          | -0.117333333 | 0.18727253  | 0.3333333    | 0.55555556   |
| DBS Bank  | 2024-01-06 | 1          | 0.250000000  | NA          | 1.0000000    | 0.00000000   |
| OCBC Bank | 2024-01-06 | 1          | 0.202000000  | NA          | 1.0000000    | 0.00000000   |
| UOB Bank  | 2024-01-06 | 1          | 0.202000000  | NA          | 1.0000000    | 0.00000000   |
| DBS Bank  | 2024-01-07 | 2          | -0.125000000 | 0.17677670  | 0.0000000    | 0.50000000   |
| OCBC Bank | 2024-01-07 | 2          | 0.000000000  | 0.00000000  | 0.0000000    | 0.00000000   |



## Sentiment data

### **Link Positive, Negative and Neutral Sentiment to Price Movements**

We examine whether strong positive, negative or neutral sentiment precedes measurable changes in returns, volatility, or direction.

### **Integrate Sentiment to Improve Prediction Accuracy**

Sentiment features are added to our models to enhance forecasting performance, especially during event-heavy or high-volatility periods.



## Tools and Technologies

### **Python:**

yFinance API, TensorFlow, Keras, scikit-learn, pandas, numpy, matplotlib and seaborn

### **R code:**

GDELT API

stats → for lm() (Linear Regression)

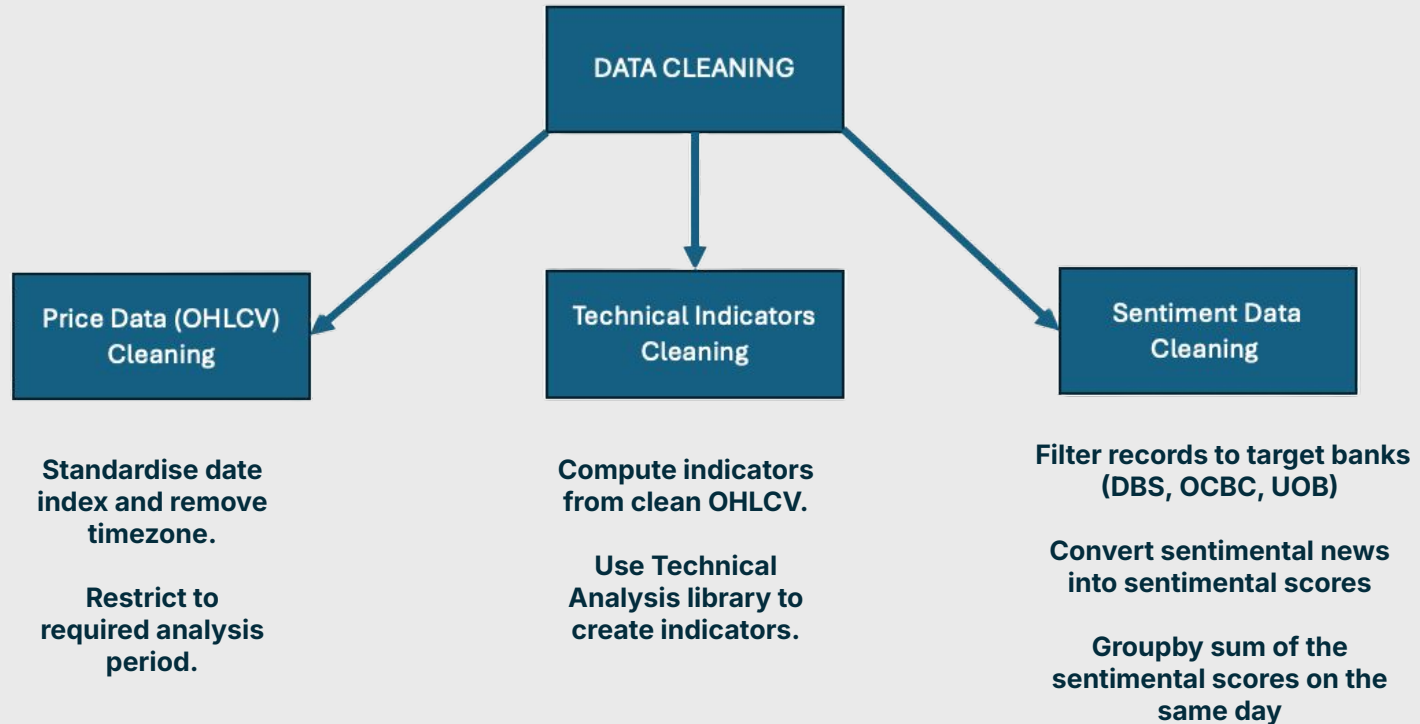
glmnet → for LASSO (alpha = 1) and Ridge (alpha = 0)

### **Power BI:** Dashboard

# METHODOLOGY



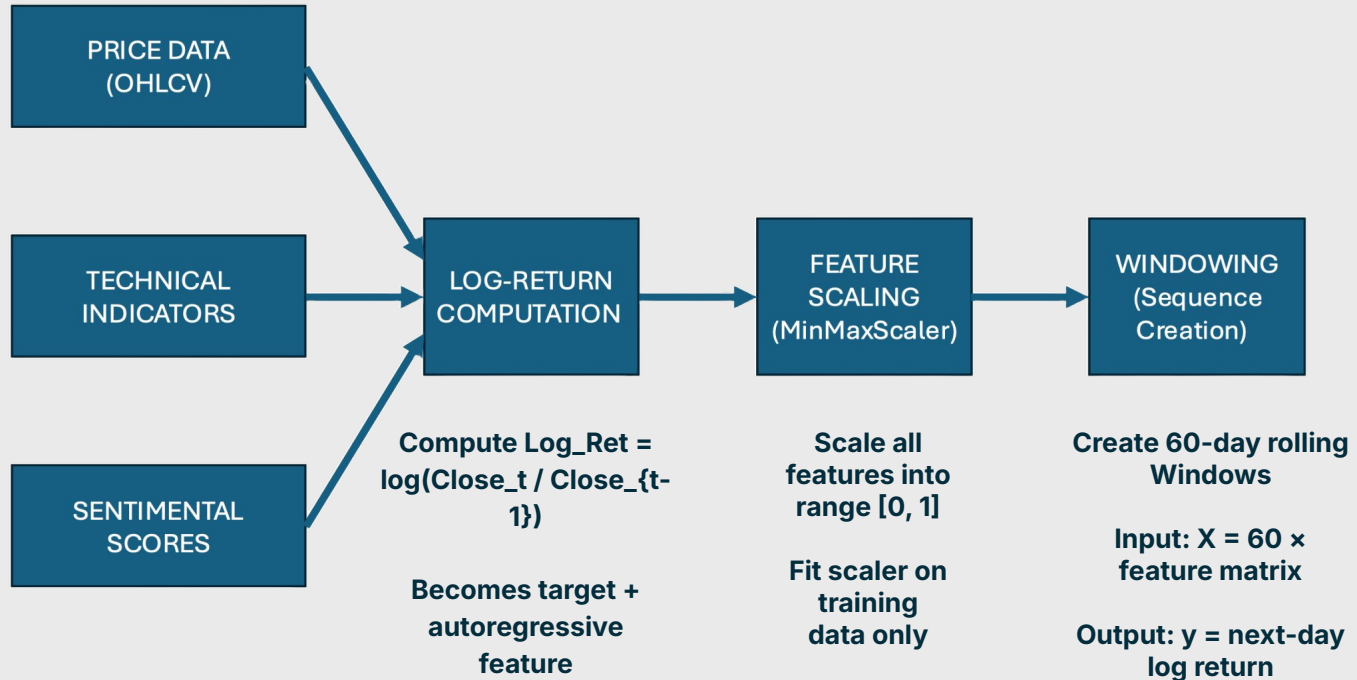
# Data Cleaning





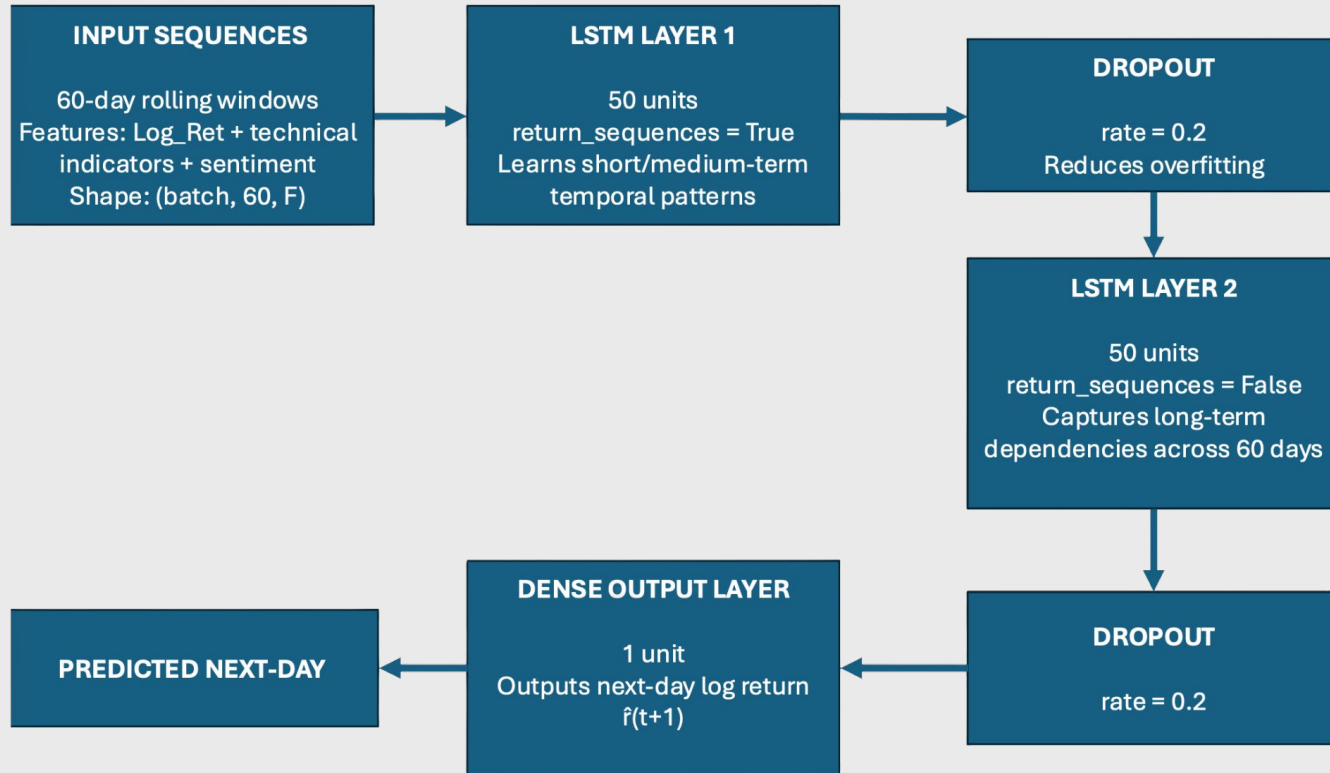
# Data Transformation

## MERGING DATASETS

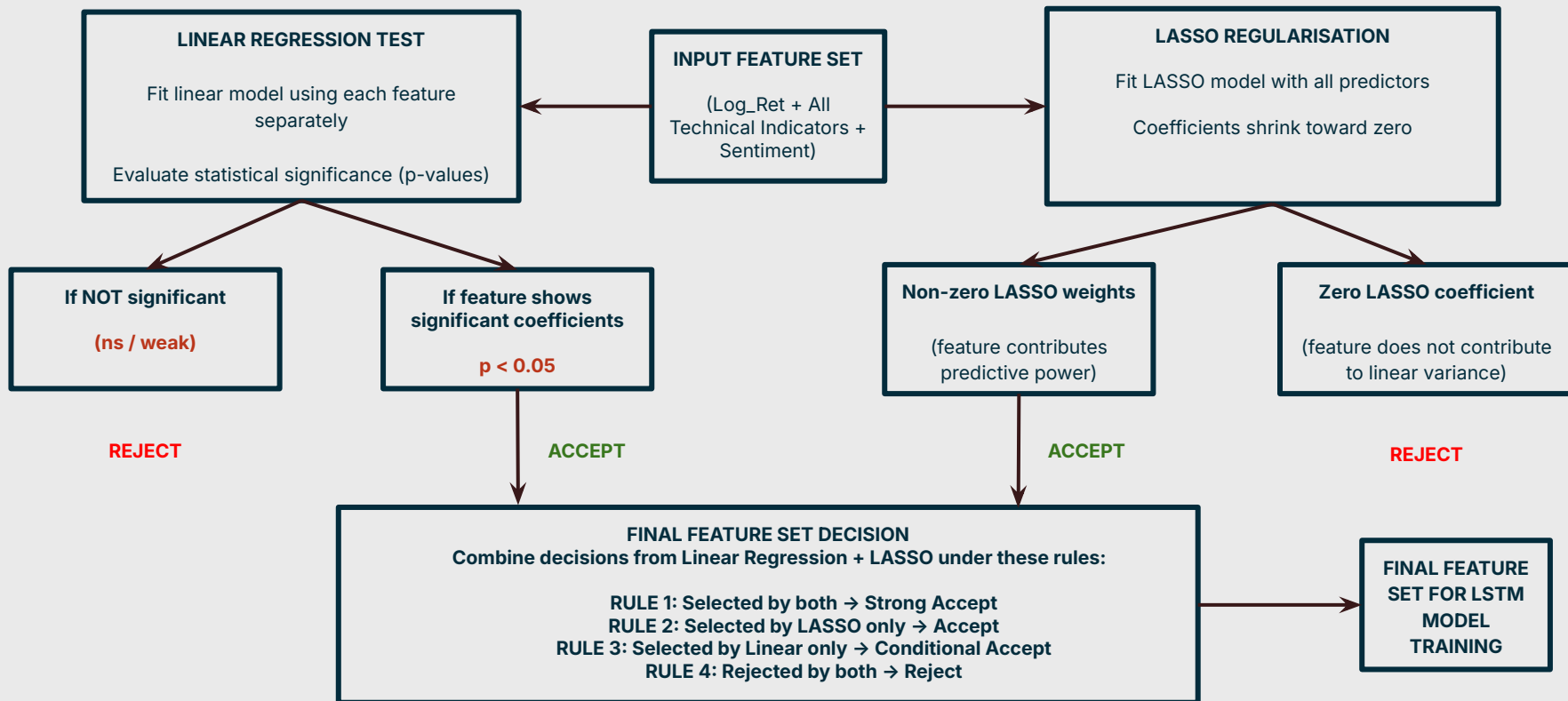




## Modeling and Design : LSTM architecture



## → Modeling and Design : Feature Selection





# Testing Techniques: Walk Forward Expanding Window Validation

## STEP 1. SET PREDICTION START DATE

SELECT START DATE FOR PREDICTION WINDOW

- Example: Start at 2025-01-01

## STEP 2. TRAIN LSTM USING EXPANDING WINDOW

Training range: 2024-01-01 → Current walk-forward date 2025-01-01

Uses 60-day sequences of scaled features

Model learns temporal patterns

## STEP 3. GENERATE 1-DAY-AHEAD FORECAST

Predict next-day log return

Convert log return → price estimate:

$$P_{\text{pred}}(t+1) = P(t) \times \exp(\hat{r}(t+1))$$

## STEP 4. EVALUATE DIRECTIONAL ACCURACY

Compare predicted vs actual movement relative to the last known price before test window

If both move UP or both move DOWN → Correct

## STEP 5. SHIFT WINDOW FORWARD BY 1 WEEK

Move prediction date +7 days

Expanding training window increases in size

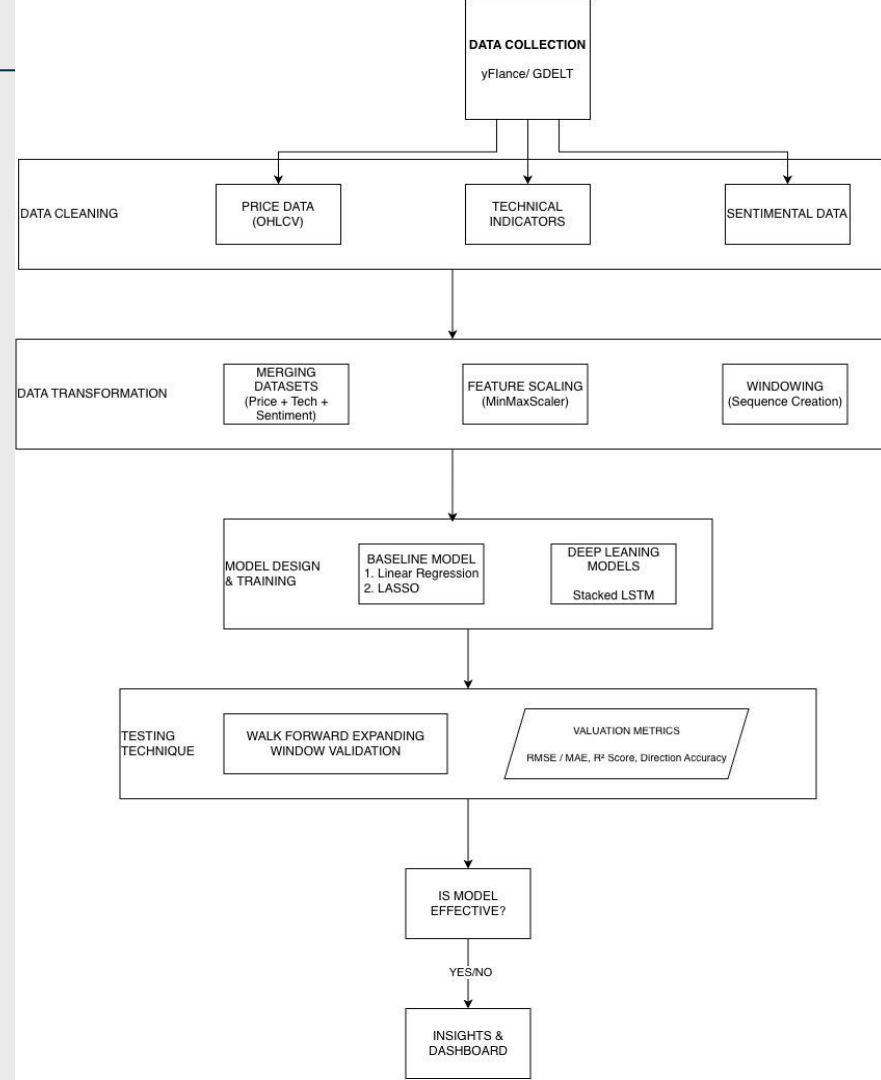
Repeat training + prediction + evaluation

## STEP 6. REPEAT UNTIL END OF TEST PERIOD

Compute overall average performance



## Methodology Flow



# FINDING & RESULTS

# Feature selection: Linear regression

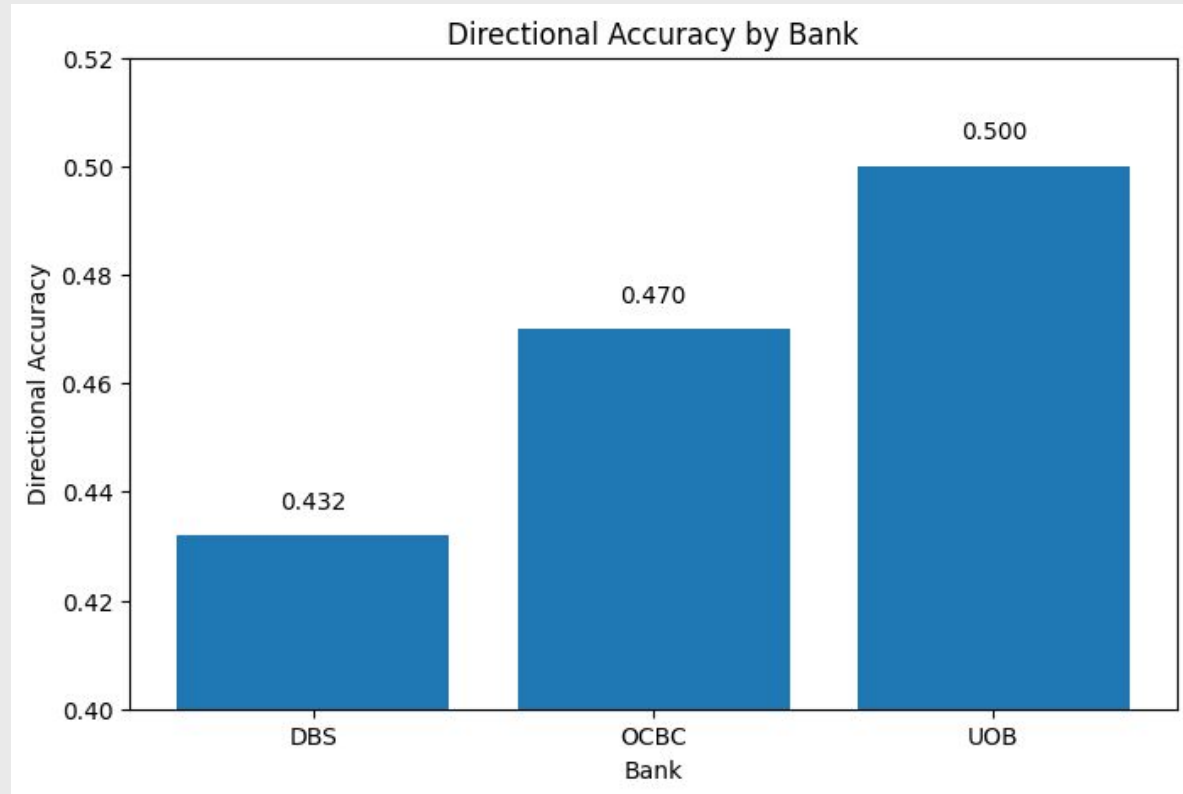
| Indicator                         | Type            | UOB | DBS | OCBC |
|-----------------------------------|-----------------|-----|-----|------|
| Rate of Change                    | Momentum        | *** | *** | **   |
| Triple Exponential Moving Average | Trend (Custom)  | ns  | *** | ***  |
| Bollinger Band Upper              | Volatility      | *** | **  | ***  |
| Momentum indicator                | Department name | ns  | *** | ***  |
| MACD Histogram                    | Trend           | .   | *** | **   |
| Exponential Moving Average        | Trend           | .   | *** | ***  |
| Simple Moving Average             | Trend           | ns  | *** | **   |

## Feature selection: LASSO

| Bank | Features Selected (non-zero) |
|------|------------------------------|
| OCBC | MOM, ROC                     |
| DBS  | MOM, ROC                     |
| UOB  | MOM, ROC                     |



## Model accuracy





# UOB Performance

<https://www.straitstimes.com/business/banking/uob-q3-profit-tumbles-72-to-443-million-as-bank-raises-pre-emptive-provisions>

THE STRAITS TIMES

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## UOB third-quarter profit tumbles 72% as bank hikes allowances by \$1 billion

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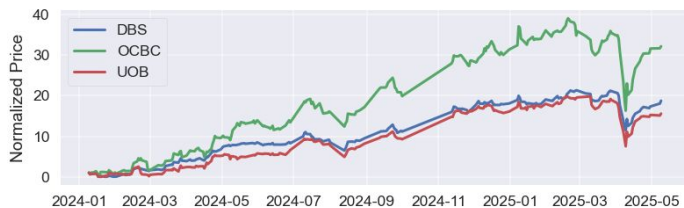
UOB noted that its dividend payment would not be affected by the pre-emptive general allowance.

ST PHOTO: GIN TAY

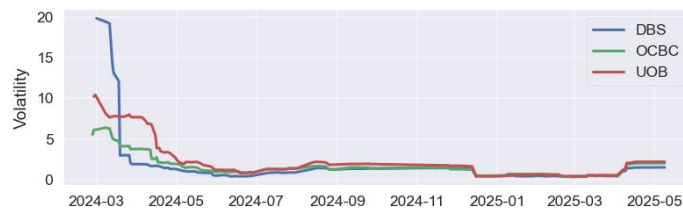


# Visualizations

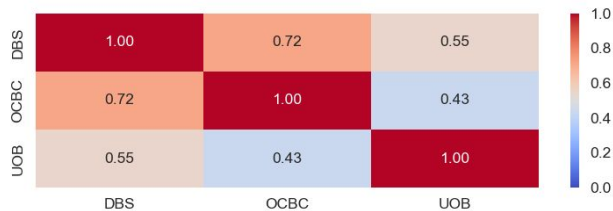
Normalized Stock Price Trends



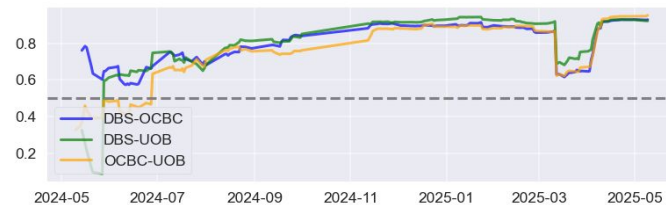
21-Day Rolling Volatility (Annualized)



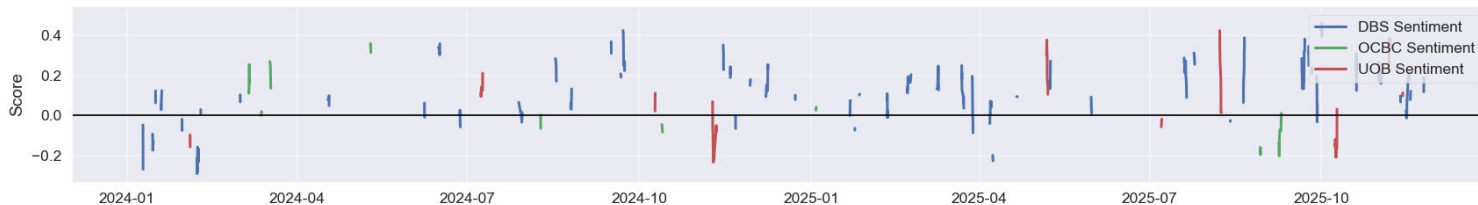
Correlation Matrix of Returns



60-Day Rolling Correlations



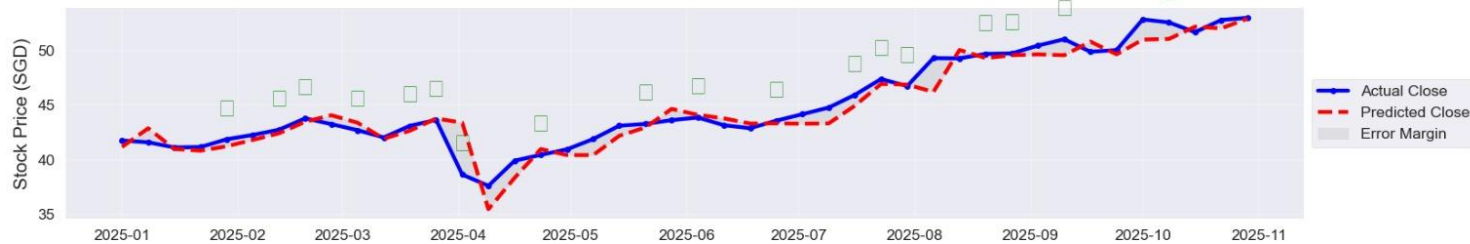
News Sentiment Score (10-Day Smoothed)



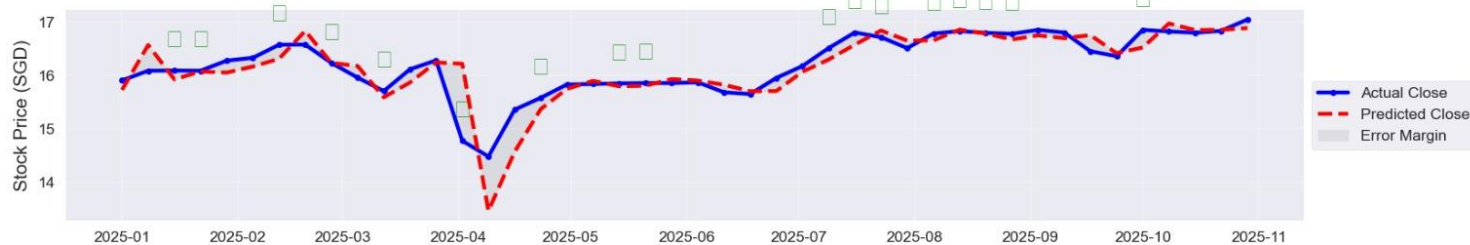


# Visualizations

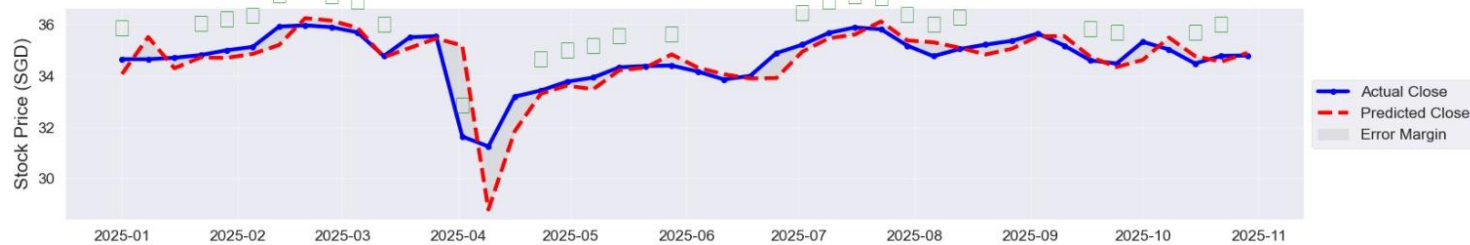
**DBS: Actual vs. Predicted Price Trend**



**OCBC: Actual vs. Predicted Price Trend**



**UOB: Actual vs. Predicted Price Trend**





## Strengths of the approach

- Combines Multiple Sources of Market Information
- Uses Log Returns: More Stable and Theoretically Correct
- LSTM Captures Nonlinear and Temporal Dependencies.
- Walk-Forward Expanding-Window Testing Removes Look-Ahead Bias.
- Directional Accuracy Matches Real Trading Goals.
- Sentiment Helps Even When Linear Models Cannot Detect It.



## Limitations of approach

- Accuracy ranging from 0.43 to 0.5
- Difficult to predict noise filled financial data.
- LSTM model affected by sensitive market regimes.
- Sentimental analysis being significant to show that there are nonlinear effects that are difficult to interpret.
- Unable to capture the intraday shifts.
- One day predictions cause challenges when there are sudden market changes.



## Future improvements

- Introducing LSTM/GRU deep learning extended architectures
- Exploring traditional ML models such as XGBoost and Random Forest
- Integrating more macroeconomic variables (GDP, interest rates, regional indices)
- Adding global banking indices for comparative analysis

Q&A